BTS (C) – V – (S) – 10 – 035 – L

B. Tech Degree V Semester (Supplementary) Examination July 2010

ME 503 MECHANICS OF MACHINERY

(2006 Scheme)

Time : 3 Hours Maximum Marks : 100

PART - A

(Answer ***ALL*** questions)

(8 x 5 = 40)

I. (a) What is meant by degree of freedom of a mechanism? How is it determined?

(b) For the slider crank mechanism, the lengths of crank and connecting rod are 200 mm and 800 mm respectively. Locate all the I-centres for the position of the crank when it has turned 30° from the IDC.

(c) Differentiate between Direct Kinematics and Inverse Kinematics applied to robot arm synthesis.

(d) Discuss with the help of sketches Overlay method of linkage synthesis.

(e) Explain the concept of Interference in gears.

(f) State and derive Law of Gearing.

(g) A screw jack is required to take a load of 20 kN. If the pitch and diameter of the screw are 8 mm and 40 mm respectively, determine the ratio of torques required to raise and lower the load. Assume that load rotates with screw. Take μ between screw and nut as 0.1.

(h) What are the requirements of a high speed cam? Justify.

PART - B

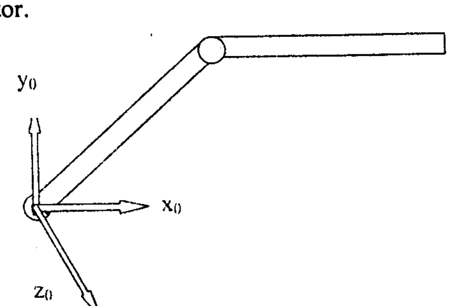
(4 x 15 = 60)

II. Discuss with the help of sketches, the functioning of any one quick-return mechanism. Where are they used? Give examples. (15)

OR

III. The following data applies for a four-link mechanism. Link AD = 100 mm (fixed), Link AB = 50 mm, Planar Link BFC $\rightarrow$ BF = 45 mm, BC = 66 mm and CF = 30 mm, Planar Link DGC $\rightarrow$ DC = 56 mm, DG = 44 mm and CG = 24 mm, Angle BAD = 60°. Point F lies within the loop ABCD where as G is outside. The link AB has an instantaneous angular velocity of 10.5 rad/s clockwise and a retardation of 26 rad/s^2 in counter clockwise direction. Find the linear accelerations of points F and G. (15)

IV. A two degree freedom manipulator is shown below. The length of each link is 1 m, establish its link coordinate frames and find ${}^0A_1$ and ${}^1A_2$ Find the inverse kinematic solution for this manipulator. (15)



(P.T.O.)

OR

V. Apply Freudensteins method to design a four link mechanism. Given three positions of input and output links as

$$\theta_1 = 20^0 \qquad \Phi_1 = 35^0$$
$$\theta_2 = 35^0 \qquad \Phi_2 = 45^0$$
$$\theta_3 = 50^0 \qquad \Phi_3 = 60^0$$

(15)

VI. The following data refer to two meshing gears having 20° involute teeth. Number of teeth on wheel = 52, Number of teeth on pinion = 20, Speed of pinion = 360 rpm, Module = 8 mm. If the addendum of each gear is such that the path of approach and path of recess are half of their maximum possible values, determine the addendum for gear and pinion and also length of arc of contact. (15)

OR

VII. Determine a suitable train of wheels to satisfy the requirements of a clock, the minute hand of which is fixed to a spindle and the hour hand to a sleeve rotating freely on the same spindle. The pitch is same for all the wheels and each wheel has at least 11 teeth. Total number of teeth should be as small as possible. (15)

VIII. The push rod of a valve of an internal combustion engine ascends with uniform acceleration and retardation, along a path inclined to the vertical by 60°, descends with simple harmonic motion. The base circle diameter of cam is 50 cm, and push rod has a roller of diameter 16 mm fitted to its end. The axis of roller and cam lie on the same vertical straight line. The maximum stroke of the follower is 2 cm in the direction inclined at 60° to the vertical. The angle of action for outstroke and return stroke is 60° each interposed by a dwell period of 60°. Draw the profile of the cam. (15)

OR

IX. A turn buckle having right and lelft hand threads is used to couple two railway coaches. It has single start square threads with pitch of 1.25 cm on a mean diameter of 3.75 cm. Taking $\mu = 0.1$, find the work done in drawing the coaches together a distance of 20 cm :

(i) against a steady load of 2945 N
(ii) if the load increases over 20 cm from 2940 N to 9800 N
(iii) efficiency of the turnbuckle. (15)

***